

Newman-Penrose Formalism Calculations

Stephani et al., Equation (35.30_e1)

Metric

$$ds^2 = -2 \left(A - \frac{b_x^2 + b_y^2}{P} \right) du^2 - 2 du dv + \frac{4b_x}{P} du dx + \frac{4b_y}{P} du dy + \frac{2}{P} (dx^2 + dy^2).$$

Coordinates

$$(x^\mu) = (u, v, x, y)$$

Metric Functions

$$K = K(u),$$

$$H = H(u, x, y),$$

$$Q = 1 + x^2 + y^2,$$

$$P = k^2 Q^2,$$

$$A = v^2 k^2 + lv + H,$$

$$b_x = b_x(u, x, y),$$

$$b_y = b_y(u, x, y).$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(v^2 k^2 + lv - \frac{b_x(u, x, y)^2}{(x^4 + 2x^2 y^2 + y^4 + 2x^2 + 2y^2 + 1)k^2} - \frac{b_y(u, x, y)^2}{(x^4 + 2x^2 y^2 + y^4 + 2x^2 + 2y^2 + 1)k^2} + H \right) du + (1) dv \\ + \left(-\frac{2b_x(u, x, y)}{(x^4 + 2x^2 y^2 + y^4 + 2x^2 + 2y^2 + 1)k^2} \right) dx + \left(-\frac{2b_y(u, x, y)}{(x^4 + 2x^2 y^2 + y^4 + 2x^2 + 2y^2 + 1)k^2} \right) dy$$

$$m = \left(\frac{1}{(x^2 + y^2 + 1)k} \right) dx + \left(\frac{i}{(x^2 + y^2 + 1)k} \right) dy$$

$$\bar{m} = \left(\frac{1}{(x^2 + y^2 + 1)k} \right) dx + \left(-\frac{i}{(x^2 + y^2 + 1)k} \right) dy$$

Contravariant components

$$l = (-1) dv$$

$$n = (-1) du + \left(\frac{v^2 x^4 k^4}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{2 v^2 x^2 y^2 k^4}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{v^2 y^4 k^4}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} \right. \\ + \frac{lv x^4 k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{2 lv x^2 y^2 k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{lv y^4 k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} \\ + \frac{2 v^2 x^2 k^4}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{2 v^2 y^2 k^4}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{x^4 H k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} \\ + \frac{2 x^2 y^2 H k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{y^4 H k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{2 lv x^2 k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} \\ + \frac{2 lv y^2 k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{v^2 k^4}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{2 x^2 H k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} \\ + \frac{2 y^2 H k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{lv k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{H k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} \\ \left. - \frac{b_x(u, x, y)^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} - \frac{b_y(u, x, y)^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} \right) dv$$

$$m = \left(\frac{b_x(u, x, y)}{x^2 k + y^2 k + k} + \frac{i b_y(u, x, y)}{x^2 k + y^2 k + k} \right) dv + \left(\frac{1}{2} x^2 k + \frac{1}{2} y^2 k + \frac{1}{2} k \right) dx + \left(\frac{1}{2} i x^2 k + \frac{1}{2} i y^2 k + \frac{1}{2} i k \right) dy$$

$$\bar{m} = \left(\frac{b_x(u, x, y)}{x^2 k + y^2 k + k} - \frac{i b_y(u, x, y)}{x^2 k + y^2 k + k} \right) dv + \left(\frac{1}{2} x^2 k + \frac{1}{2} y^2 k + \frac{1}{2} k \right) dx + \left(-\frac{1}{2} i x^2 k - \frac{1}{2} i y^2 k - \frac{1}{2} i k \right) dy$$

Directional Derivatives

$$D = -\partial_v$$

$$\begin{aligned} \Delta = & -\partial_u + \frac{v^2 x^4 k^4}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{2 v^2 x^2 y^2 k^4}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{v^2 y^4 k^4}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} \\ & + \frac{lv x^4 k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{2 lv x^2 y^2 k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{lv y^4 k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} \\ & + \frac{2 v^2 x^2 k^4}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{2 v^2 y^2 k^4}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{x^4 H k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} \\ & + \frac{2 x^2 y^2 H k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{y^4 H k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{2 lv x^2 k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} \\ & + \frac{2 lv y^2 k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{v^2 k^4}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{2 x^2 H k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} \\ & + \frac{2 y^2 H k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{lv k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} + \frac{H k^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} \\ & - \frac{b_x(u, x, y)^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} - \frac{b_y(u, x, y)^2}{x^4 k^2 + 2 x^2 y^2 k^2 + y^4 k^2 + 2 x^2 k^2 + 2 y^2 k^2 + k^2} \partial_v \end{aligned}$$

$$\delta = \frac{b_x(u, x, y)}{x^2 k + y^2 k + k} + \frac{i b_y(u, x, y)}{x^2 k + y^2 k + k} \partial_v + \frac{1}{2} x^2 k + \frac{1}{2} y^2 k + \frac{1}{2} k \partial_x + \frac{1}{2} i x^2 k + \frac{1}{2} i y^2 k + \frac{1}{2} i k \partial_y$$

$$\bar{\delta} = \frac{b_x(u, x, y)}{x^2 k + y^2 k + k} - \frac{i b_y(u, x, y)}{x^2 k + y^2 k + k} \partial_v + \frac{1}{2} x^2 k + \frac{1}{2} y^2 k + \frac{1}{2} k \partial_x - \frac{1}{2} i x^2 k - \frac{1}{2} i y^2 k - \frac{1}{2} i k \partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\begin{aligned}\mu = & -\frac{i x^2 k \frac{\partial}{\partial y} b_x(u, x, y)}{2(x^2 k + y^2 k + k)} - \frac{i y^2 k \frac{\partial}{\partial y} b_x(u, x, y)}{2(x^2 k + y^2 k + k)} + \frac{i x^2 k \frac{\partial}{\partial x} b_y(u, x, y)}{2(x^2 k + y^2 k + k)} + \frac{i y^2 k \frac{\partial}{\partial x} b_y(u, x, y)}{2(x^2 k + y^2 k + k)} + \frac{2i y b_x(u, x, y) k}{x^2 k + y^2 k + k} \\ & - \frac{2i x b_y(u, x, y) k}{x^2 k + y^2 k + k} + \frac{x^2 \frac{\partial}{\partial u} k}{x^2 k + y^2 k + k} + \frac{y^2 \frac{\partial}{\partial u} k}{x^2 k + y^2 k + k} - \frac{i k \frac{\partial}{\partial y} b_x(u, x, y)}{2(x^2 k + y^2 k + k)} + \frac{i k \frac{\partial}{\partial x} b_y(u, x, y)}{2(x^2 k + y^2 k + k)} + \frac{\frac{\partial}{\partial u} k}{x^2 k + y^2 k + k}\end{aligned}$$

$$\lambda = 0$$

$$\begin{aligned}\nu = & \frac{x^6 k^3 \frac{\partial}{\partial x} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} + \frac{3x^4 y^2 k^3 \frac{\partial}{\partial x} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} + \frac{3x^2 y^4 k^3 \frac{\partial}{\partial x} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} \\ & + \frac{y^6 k^3 \frac{\partial}{\partial x} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} - \frac{i x^6 k^3 \frac{\partial}{\partial y} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} - \frac{3i x^4 y^2 k^3 \frac{\partial}{\partial y} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} \\ & - \frac{3i x^2 y^4 k^3 \frac{\partial}{\partial y} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} - \frac{i y^6 k^3 \frac{\partial}{\partial y} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} + \frac{3x^4 k^3 \frac{\partial}{\partial x} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} \\ & + \frac{3x^2 y^2 k^3 \frac{\partial}{\partial x} H}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \frac{3y^4 k^3 \frac{\partial}{\partial x} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} - \frac{3i x^4 k^3 \frac{\partial}{\partial y} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} \\ & - \frac{3i x^2 y^2 k^3 \frac{\partial}{\partial y} H}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} - \frac{3i y^4 k^3 \frac{\partial}{\partial y} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} + \frac{2v x^2 b_x(u, x, y) k^3}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} \\ & + \frac{2v y^2 b_x(u, x, y) k^3}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} - \frac{2i v x^2 b_y(u, x, y) k^3}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} - \frac{2i v y^2 b_y(u, x, y) k^3}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} \\ & + \frac{3x^2 k^3 \frac{\partial}{\partial x} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} + \frac{3y^2 k^3 \frac{\partial}{\partial x} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} - \frac{3i x^2 k^3 \frac{\partial}{\partial y} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} \\ & - \frac{3i y^2 k^3 \frac{\partial}{\partial y} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} + \frac{l x^2 b_x(u, x, y) k}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \frac{l y^2 b_x(u, x, y) k}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} \\ & - \frac{i l x^2 b_y(u, x, y) k}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} - \frac{i l y^2 b_y(u, x, y) k}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \frac{2v b_x(u, x, y) k^3}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} \\ & - \frac{2i v b_y(u, x, y) k^3}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} - \frac{x^2 b_x(u, x, y) k \frac{\partial}{\partial x} b_x(u, x, y)}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} - \frac{y^2 b_x(u, x, y) k \frac{\partial}{\partial x} b_x(u, x, y)}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \dots\end{aligned}$$

$$\alpha = \frac{1}{2} x k - \frac{1}{2} i y k$$

$$\beta = -\frac{1}{2} x k - \frac{1}{2} i y k$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\begin{aligned} \gamma = & -\frac{vx^2k^2}{x^2+y^2+1} - \frac{vy^2k^2}{x^2+y^2+1} - \frac{lx^2}{2(x^2+y^2+1)} - \frac{ly^2}{2(x^2+y^2+1)} - \frac{vk^2}{x^2+y^2+1} - \frac{ix^2\frac{\partial}{\partial y}b_x(u,x,y)}{4(x^2+y^2+1)} - \frac{iy^2\frac{\partial}{\partial y}b_x(u,x,y)}{4(x^2+y^2+1)} \\ & + \frac{ix^2\frac{\partial}{\partial x}b_y(u,x,y)}{4(x^2+y^2+1)} + \frac{iy^2\frac{\partial}{\partial x}b_y(u,x,y)}{4(x^2+y^2+1)} + \frac{iyb_x(u,x,y)}{x^2+y^2+1} - \frac{ixb_y(u,x,y)}{x^2+y^2+1} - \frac{l}{2(x^2+y^2+1)} - \frac{i\frac{\partial}{\partial y}b_x(u,x,y)}{4(x^2+y^2+1)} + \frac{i\frac{\partial}{\partial x}b_y(u,x,y)}{4(x^2+y^2+1)} \end{aligned}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = k^2$$

$$\begin{aligned}
\Phi_{22} = & \frac{x^8 k^4 \frac{\partial^2}{(\partial x)^2} H}{4(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} + \frac{x^6 y^2 k^4 \frac{\partial^2}{(\partial x)^2} H}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \frac{3x^4 y^4 k^4 \frac{\partial^2}{(\partial x)^2} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} \\
& + \frac{x^2 y^6 k^4 \frac{\partial^2}{(\partial x)^2} H}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \frac{y^8 k^4 \frac{\partial^2}{(\partial x)^2} H}{4(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} + \frac{x^8 k^4 \frac{\partial^2}{(\partial y)^2} H}{4(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} \\
& + \frac{x^6 y^2 k^4 \frac{\partial^2}{(\partial y)^2} H}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \frac{3x^4 y^4 k^4 \frac{\partial^2}{(\partial y)^2} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} + \frac{x^2 y^6 k^4 \frac{\partial^2}{(\partial y)^2} H}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} \\
& + \frac{y^8 k^4 \frac{\partial^2}{(\partial y)^2} H}{4(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} + \frac{x^6 k^4 \frac{\partial^2}{(\partial x)^2} H}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \frac{3x^4 y^2 k^4 \frac{\partial^2}{(\partial x)^2} H}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} \\
& + \frac{3x^2 y^4 k^4 \frac{\partial^2}{(\partial x)^2} H}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \frac{y^6 k^4 \frac{\partial^2}{(\partial x)^2} H}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \frac{x^6 k^4 \frac{\partial^2}{(\partial y)^2} H}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} \\
& + \frac{3x^4 y^2 k^4 \frac{\partial^2}{(\partial y)^2} H}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \frac{3x^2 y^4 k^4 \frac{\partial^2}{(\partial y)^2} H}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \frac{y^6 k^4 \frac{\partial^2}{(\partial y)^2} H}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} \\
& + \frac{vx^4 k^4 \frac{\partial}{\partial x} b_x(u, x, y)}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \frac{2vx^2 y^2 k^4 \frac{\partial}{\partial x} b_x(u, x, y)}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \frac{vy^4 k^4 \frac{\partial}{\partial x} b_x(u, x, y)}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} \\
& + \frac{vx^4 k^4 \frac{\partial}{\partial y} b_y(u, x, y)}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \frac{2vx^2 y^2 k^4 \frac{\partial}{\partial y} b_y(u, x, y)}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \frac{vy^4 k^4 \frac{\partial}{\partial y} b_y(u, x, y)}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} \\
& - \frac{4vx^3 b_x(u, x, y) k^4}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} - \frac{4vxy^2 b_x(u, x, y) k^4}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} - \frac{4vx^2 y b_y(u, x, y) k^4}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} \\
& - \frac{4vy^3 b_y(u, x, y) k^4}{x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2} + \frac{3x^4 k^4 \frac{\partial^2}{(\partial x)^2} H}{2(x^4 k^2 + 2x^2 y^2 k^2 + y^4 k^2 + 2x^2 k^2 + 2y^2 k^2 + k^2)} + \dots
\end{aligned}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\begin{aligned}
\Psi_3 = & -\frac{i x^4 k \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{8(x^2 + y^2 + 1)} - \frac{i x^2 y^2 k \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{4(x^2 + y^2 + 1)} - \frac{i y^4 k \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{8(x^2 + y^2 + 1)} - \frac{x^4 k \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{8(x^2 + y^2 + 1)} - \frac{x^2 y^2 k \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{4(x^2 + y^2 + 1)} - \frac{y^4 k \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{8(x^2 + y^2 + 1)} + \frac{i x^4 k \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{8(x^2 + y^2 + 1)} \\
& + \frac{i x^2 y^2 k \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{4(x^2 + y^2 + 1)} + \frac{i y^4 k \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{8(x^2 + y^2 + 1)} + \frac{x^4 k \frac{\partial^2}{\partial x \partial y} b_y(u, x, y)}{8(x^2 + y^2 + 1)} + \frac{x^2 y^2 k \frac{\partial^2}{\partial x \partial y} b_y(u, x, y)}{4(x^2 + y^2 + 1)} + \frac{y^4 k \frac{\partial^2}{\partial x \partial y} b_y(u, x, y)}{8(x^2 + y^2 + 1)} + \frac{i x^2 y k \frac{\partial}{\partial x} b_x(u, x, y)}{2(x^2 + y^2 + 1)} + \frac{i y^3 k \frac{\partial}{\partial x} b_x(u, x, y)}{2(x^2 + y^2 + 1)} \\
& + \frac{x^2 y k \frac{\partial}{\partial y} b_x(u, x, y)}{2(x^2 + y^2 + 1)} + \frac{y^3 k \frac{\partial}{\partial y} b_x(u, x, y)}{2(x^2 + y^2 + 1)} - \frac{i x^3 k \frac{\partial}{\partial x} b_y(u, x, y)}{2(x^2 + y^2 + 1)} - \frac{i x y^2 k \frac{\partial}{\partial x} b_y(u, x, y)}{2(x^2 + y^2 + 1)} - \frac{x^3 k \frac{\partial}{\partial y} b_y(u, x, y)}{2(x^2 + y^2 + 1)} - \frac{x y^2 k \frac{\partial}{\partial y} b_y(u, x, y)}{2(x^2 + y^2 + 1)} + \frac{x^2 b_x(u, x, y) k}{2(x^2 + y^2 + 1)} \\
& - \frac{i x y b_x(u, x, y) k}{x^2 + y^2 + 1} - \frac{y^2 b_x(u, x, y) k}{2(x^2 + y^2 + 1)} + \frac{i x^2 b_y(u, x, y) k}{2(x^2 + y^2 + 1)} + \frac{x y b_y(u, x, y) k}{x^2 + y^2 + 1} - \frac{i y^2 b_y(u, x, y) k}{2(x^2 + y^2 + 1)} - \frac{i x^2 k \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{4(x^2 + y^2 + 1)} - \frac{i y^2 k \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{4(x^2 + y^2 + 1)} - \frac{x^2 k \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{4(x^2 + y^2 + 1)} \\
& - \frac{y^2 k \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{4(x^2 + y^2 + 1)} + \frac{i x^2 k \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{4(x^2 + y^2 + 1)} + \frac{i y^2 k \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{4(x^2 + y^2 + 1)} + \frac{x^2 k \frac{\partial^2}{\partial x \partial y} b_y(u, x, y)}{4(x^2 + y^2 + 1)} + \frac{y^2 k \frac{\partial^2}{\partial x \partial y} b_y(u, x, y)}{4(x^2 + y^2 + 1)} + \frac{i y k \frac{\partial}{\partial x} b_x(u, x, y)}{2(x^2 + y^2 + 1)} + \frac{y k \frac{\partial}{\partial y} b_x(u, x, y)}{2(x^2 + y^2 + 1)} \\
& - \frac{i x k \frac{\partial}{\partial x} b_y(u, x, y)}{2(x^2 + y^2 + 1)} - \frac{x k \frac{\partial}{\partial y} b_y(u, x, y)}{2(x^2 + y^2 + 1)} - \frac{b_x(u, x, y) k}{2(x^2 + y^2 + 1)} + \frac{i b_y(u, x, y) k}{2(x^2 + y^2 + 1)} - \frac{i k \frac{\partial^2}{\partial x \partial y} b_x(u, x, y)}{8(x^2 + y^2 + 1)} - \frac{k \frac{\partial^2}{(\partial y)^2} b_x(u, x, y)}{8(x^2 + y^2 + 1)} + \frac{i k \frac{\partial^2}{(\partial x)^2} b_y(u, x, y)}{8(x^2 + y^2 + 1)} + \frac{k \frac{\partial^2}{\partial x \partial y} b_y(u, x, y)}{8(x^2 + y^2 + 1)}
\end{aligned}$$

$$\begin{aligned}
\Psi_4 = & \frac{x^8 k^3 \frac{\partial^2}{(\partial x)^2} H}{4(x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k)} + \frac{x^6 y^2 k^3 \frac{\partial^2}{(\partial x)^2} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} + \frac{3x^4 y^4 k^3 \frac{\partial^2}{(\partial x)^2} H}{2(x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k)} \\
& + \frac{x^2 y^6 k^3 \frac{\partial^2}{(\partial x)^2} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} + \frac{y^8 k^3 \frac{\partial^2}{(\partial x)^2} H}{4(x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k)} - \frac{i x^8 k^3 \frac{\partial^2}{\partial x \partial y} H}{2(x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k)} \\
& - \frac{2i x^6 y^2 k^3 \frac{\partial^2}{\partial x \partial y} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} - \frac{3i x^4 y^4 k^3 \frac{\partial^2}{\partial x \partial y} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} - \frac{2i x^2 y^6 k^3 \frac{\partial^2}{\partial x \partial y} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} \\
& - \frac{i y^8 k^3 \frac{\partial^2}{\partial x \partial y} H}{2(x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k)} - \frac{x^8 k^3 \frac{\partial^2}{(\partial y)^2} H}{4(x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k)} \\
& - \frac{x^6 y^2 k^3 \frac{\partial^2}{(\partial y)^2} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} - \frac{3x^4 y^4 k^3 \frac{\partial^2}{(\partial y)^2} H}{2(x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k)} - \frac{x^2 y^6 k^3 \frac{\partial^2}{(\partial y)^2} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} \\
& - \frac{y^8 k^3 \frac{\partial^2}{(\partial y)^2} H}{4(x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k)} + \frac{x^7 k^3 \frac{\partial}{\partial x} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} - \frac{i x^6 y k^3 \frac{\partial}{\partial x} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} \\
& + \frac{3x^5 y^2 k^3 \frac{\partial}{\partial x} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} - \frac{3i x^4 y^3 k^3 \frac{\partial}{\partial x} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} + \frac{3x^3 y^4 k^3 \frac{\partial}{\partial x} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} \\
& - \frac{3i x^2 y^5 k^3 \frac{\partial}{\partial x} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} + \frac{xy^6 k^3 \frac{\partial}{\partial x} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} - \frac{i y^7 k^3 \frac{\partial}{\partial x} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} \\
& - \frac{i x^7 k^3 \frac{\partial}{\partial y} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} - \frac{x^6 y k^3 \frac{\partial}{\partial y} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} - \frac{3i x^5 y^2 k^3 \frac{\partial}{\partial y} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} \\
& - \frac{3x^4 y^3 k^3 \frac{\partial}{\partial y} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} - \frac{3i x^3 y^4 k^3 \frac{\partial}{\partial y} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} - \frac{3x^2 y^5 k^3 \frac{\partial}{\partial y} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} \\
& - \frac{i xy^6 k^3 \frac{\partial}{\partial y} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} - \frac{y^7 k^3 \frac{\partial}{\partial y} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} + \frac{x^6 k^3 \frac{\partial^2}{(\partial x)^2} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} \\
& + \frac{3x^4 y^2 k^3 \frac{\partial^2}{(\partial x)^2} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} + \frac{3x^2 y^4 k^3 \frac{\partial^2}{(\partial x)^2} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} + \frac{y^6 k^3 \frac{\partial^2}{(\partial x)^2} H}{x^4 k + 2x^2 y^2 k + y^4 k + 2x^2 k + 2y^2 k + k} - \dots
\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "*III*"